# Quantum Optics Assignment: Worked Solutions

## Question 1: Photonic qubits and Bell states

### (a) Bell-basis orthogonality

Write the Bell states as

$$|\Phi_s\rangle = \frac{|00\rangle + s\,|11\rangle}{\sqrt{2}}, \tag{1}$$

$$|\Psi_s\rangle = \frac{|01\rangle + s\,|10\rangle}{\sqrt{2}}, \tag{2}$$

where $s \in \{+1, -1\}$.
Using the orthonormality of the computational basis,

$$\langle\Phi_s|\Phi_t\rangle = \frac{1}{2}\left(\langle 00|00\rangle + st\,\langle 11|11\rangle\right) \tag{3}$$

$$= \frac{1+st}{2}. \tag{4}$$

Therefore,

$$\langle\Phi_s|\Phi_t\rangle = \begin{cases} 1, & s = t, \\ 0, & s \neq t. \end{cases} \tag{5}$$

Similarly,

$$\langle\Psi_s|\Psi_t\rangle = \frac{1+st}{2}. \tag{6}$$

The states in the $\Phi$ and $\Psi$ families are mutually orthogonal because

$$\langle\Phi_s|\Psi_t\rangle = 0. \tag{7}$$

Hence,

$$\langle B_i|B_j\rangle = \delta_{ij}, \tag{8}$$

so the four Bell states form an orthonormal basis.

### (b) Local state of one photon

The two-photon density operator is

$$\rho_{AB} = |\Phi^+\rangle\langle\Phi^+| \tag{9}$$

$$= \frac{1}{2}\left(|HH\rangle\langle HH| + |HH\rangle\langle VV| + |VV\rangle\langle HH| + |VV\rangle\langle VV|\right). \tag{10}$$

To trace over photon $B$, use

$$\mathrm{Tr}_B\left(|i\rangle_A\,|j\rangle_B\,\langle k|_A\,\langle l|_B\right) = \langle l|j\rangle\,|i\rangle_A\,\langle k|_A\,. \tag{11}$$

Therefore,

$$\mathrm{Tr}_B\left(|HH\rangle\langle HH|\right) = |H\rangle\langle H|\,, \tag{12}$$

$$\mathrm{Tr}_B\left(|HH\rangle\langle VV|\right) = \langle V|H\rangle\,|H\rangle\langle V| = 0, \tag{13}$$

$$\mathrm{Tr}_B\left(|VV\rangle\langle HH|\right) = \langle H|V\rangle\,|V\rangle\langle H| = 0, \tag{14}$$

$$\mathrm{Tr}_B\left(|VV\rangle\langle VV|\right) = |V\rangle\langle V|\,. \tag{15}$$

Thus,

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}) \tag{16}$$
$$= \frac{1}{2}\left(|H\rangle\langle H| + |V\rangle\langle V|\right) \tag{17}$$
$$= \frac{I}{2}. \tag{18}$$

Photon $A$ is therefore in a maximally mixed state. For an $H/V$ measurement,

$$P(H) = \frac{1}{2}, \tag{19}$$
$$P(V) = \frac{1}{2}. \tag{20}$$

Equal probability for H and V.

**(c) Correlations in two bases**

From the definitions of $|D\rangle$ and $|A\rangle$,

$$|H\rangle = \frac{|D\rangle + |A\rangle}{\sqrt{2}}, \tag{21}$$
$$|V\rangle = \frac{|D\rangle - |A\rangle}{\sqrt{2}}. \tag{22}$$

Therefore,

$$|HH\rangle = \frac{1}{2}\left(|DD\rangle + |DA\rangle + |AD\rangle + |AA\rangle\right), \tag{23}$$
$$|VV\rangle = \frac{1}{2}\left(|DD\rangle - |DA\rangle - |AD\rangle + |AA\rangle\right). \tag{24}$$

Adding these expressions gives

$$|HH\rangle + |VV\rangle = |DD\rangle + |AA\rangle. \tag{25}$$

Hence,

$$|\Phi^+\rangle = \frac{|DD\rangle + |AA\rangle}{\sqrt{2}}. \tag{26}$$

In the $H/V$ basis, the possible outcomes are

$$HH \quad \text{or} \quad VV, \tag{27}$$

each with probability 1/2.
In the $D/A$ basis, the possible outcomes are

$$DD \quad \text{or} \quad AA, \tag{28}$$

each with probability 1/2.
The measurement outcomes are therefore perfectly correlated in both bases.

## Question 2: Superdense coding with polarisation qubits

### (a) Encoding the messages

Applying the identity operation gives

$$(I \otimes I)\,|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \tag{29}$$
$$= |\Phi^+\rangle\,. \tag{30}$$

Applying $X$ to Alice's qubit gives

$$(X \otimes I)\,|\Phi^+\rangle = \frac{|10\rangle + |01\rangle}{\sqrt{2}} \tag{31}$$
$$= |\Psi^+\rangle\,. \tag{32}$$

Applying $Z$ to Alice's qubit gives

$$(Z \otimes I)\,|\Phi^+\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} \tag{33}$$
$$= |\Phi^-\rangle\,. \tag{34}$$

For the final operation,

$$XZ\,|0\rangle = |1\rangle\,, \tag{35}$$
$$XZ\,|1\rangle = -\,|0\rangle\,. \tag{36}$$

Therefore,

$$(XZ \otimes I)\,|\Phi^+\rangle = \frac{|10\rangle - |01\rangle}{\sqrt{2}} \tag{37}$$
$$= -\,|\Psi^-\rangle\,. \tag{38}$$

The overall minus sign is a global phase and has no physical significance. One possible encoding table is therefore

| Classical message | Alice's operation | Resulting Bell state |
|---|---|---|
| 00 | $I$ | $\|\Phi^+\rangle$ |
| 01 | $X$ | $\|\Psi^+\rangle$ |
| 10 | $Z$ | $\|\Phi^-\rangle$ |
| 11 | $XZ$ | $\|\Psi^-\rangle$ |

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### (b) Information transmitted

Alice's four possible operations transform the shared state into four mutually orthogonal Bell states.
An ideal Bell-state measurement can distinguish all four possibilities.

### (c) Linear-optical Bell-state measurement

In terms of the input-mode creation operators,

$$|\Phi^{\pm}\rangle = \frac{1}{\sqrt{2}}\left(\hat{a}_H^\dagger \hat{b}_H^\dagger \pm \hat{a}_V^\dagger \hat{b}_V^\dagger\right)|\text{vac}\rangle\,, \tag{40}$$

$$|\Psi^{\pm}\rangle = \frac{1}{\sqrt{2}}\left(\hat{a}_H^\dagger \hat{b}_V^\dagger \pm \hat{a}_V^\dagger \hat{b}_H^\dagger\right)|\text{vac}\rangle\,. \tag{41}$$

For two photons with the same polarisation,

$$\hat{a}_\lambda^\dagger \hat{b}_\lambda^\dagger \longrightarrow \frac{1}{2}\left(\hat{c}_\lambda^\dagger + \hat{d}_\lambda^\dagger\right)\left(\hat{c}_\lambda^\dagger - \hat{d}_\lambda^\dagger\right) \tag{42}$$

$$= \frac{1}{2}\left[\left(\hat{c}_\lambda^\dagger\right)^2 - \left(\hat{d}_\lambda^\dagger\right)^2\right]. \tag{43}$$

It follows that

$$|\Phi^{\pm}\rangle \longrightarrow \frac{1}{2\sqrt{2}}\Big[\left(\hat{c}_H^\dagger\right)^2 - \left(\hat{d}_H^\dagger\right)^2 \pm \left(\hat{c}_V^\dagger\right)^2 \mp \left(\hat{d}_V^\dagger\right)^2\Big]|\text{vac}\rangle\,. \tag{44}$$

For the two $\Psi$ states,

$$\hat{a}_H^\dagger \hat{b}_V^\dagger + \hat{a}_V^\dagger \hat{b}_H^\dagger \longrightarrow \hat{c}_H^\dagger \hat{c}_V^\dagger - \hat{d}_H^\dagger \hat{d}_V^\dagger, \tag{45}$$

and

$$\hat{a}_H^\dagger \hat{b}_V^\dagger - \hat{a}_V^\dagger \hat{b}_H^\dagger \longrightarrow \hat{c}_V^\dagger \hat{d}_H^\dagger - \hat{c}_H^\dagger \hat{d}_V^\dagger. \tag{46}$$

Therefore,

$$|\Psi^{+}\rangle \longrightarrow \frac{1}{\sqrt{2}}\left(\hat{c}_H^\dagger \hat{c}_V^\dagger - \hat{d}_H^\dagger \hat{d}_V^\dagger\right)|\text{vac}\rangle\,, \tag{47}$$

$$|\Psi^{-}\rangle \longrightarrow \frac{1}{\sqrt{2}}\left(\hat{c}_V^\dagger \hat{d}_H^\dagger - \hat{c}_H^\dagger \hat{d}_V^\dagger\right)|\text{vac}\rangle\,. \tag{48}$$

The corresponding detection patterns are

| Input state | Detection pattern |
|---|---|
| $\|\Phi^+\rangle$ | Two photons in the same output port with the same polarisation |
| $\|\Phi^-\rangle$ | Two photons in the same output port with the same polarisation |
| $\|\Psi^+\rangle$ | Two orthogonally polarised photons in the same output port |
| $\|\Psi^-\rangle$ | One photon in each output port, with orthogonal polarisations |

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The states $|\Psi^+\rangle$ and $|\Psi^-\rangle$ produce different detection patterns and can therefore be distinguished unambiguously.
The states $|\Phi^+\rangle$ and $|\Phi^-\rangle$ differ only through relative phases that are not detected by photon counting. They therefore remain indistinguishable using this measurement.

### (d) Decoding efficiency

Two of the four Bell states can be identified unambiguously. For four equally likely messages, the success probability is

$$P_{\text{success}} = \frac{2}{4} \tag{50}$$

$$= \frac{1}{2}. \tag{51}$$

Thus, the standard passive linear-optical Bell-state measurement succeeds with probability 50%.
The other 50% of the events correspond to $|\Phi^+\rangle$ or $|\Phi^-\rangle$, which cannot be distinguished. Consequently, this measurement alone cannot implement deterministic dense coding.

## Question 3: BB84 quantum key distribution – Solutions

### (a) State preparation

Alice's encoding is

| Basis | Bit value | State sent |
|---|---|---|
| $Z$ | 0 | $\lvert H\rangle$ |
| $Z$ | 1 | $\lvert V\rangle$ |
| $X$ | 0 | $\lvert D\rangle = \dfrac{\lvert H\rangle + \lvert V\rangle}{\sqrt{2}}$ |
| $X$ | 1 | $\lvert A\rangle = \dfrac{\lvert H\rangle - \lvert V\rangle}{\sqrt{2}}$ |

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### (b) Basis sifting and loss

Alice and Bob each choose the $Z$ or $X$ basis with probability 1/2. Therefore, the probability that they choose the same basis is

$$P(\text{same basis}) = P(Z,Z) + P(X,X) \tag{53}$$
$$= \frac{1}{2}\frac{1}{2} + \frac{1}{2}\frac{1}{2} \tag{54}$$
$$= \frac{1}{2}. \tag{55}$$

Thus, one half of Bob's detections are retained.
The probability that a transmitted photon reaches Bob and is detected is

$$P(\text{detection}) = \eta_{\text{ch}}\eta_d. \tag{56}$$

Hence, the expected number of detections is

$$N_{\text{det}} = N\eta_{\text{ch}}\eta_d. \tag{57}$$

After basis sifting, the expected sifted-key length is

$$N_{\text{sift}} = \frac{1}{2}N\eta_{\text{ch}}\eta_d. \tag{58}$$

### (c) Intercept–resend attack

Eve chooses the same basis as Alice with probability

$$P(\text{Eve uses Alice's basis}) = \frac{1}{2}. \tag{59}$$

In this case, Eve obtains Alice's bit correctly and sends Bob the correct state. Therefore,

$$P(\text{error} \mid \text{Eve uses Alice's basis}) = 0. \tag{60}$$

Eve chooses a different basis from Alice with probability

$$P(\text{Eve uses a different basis}) = \frac{1}{2}. \tag{61}$$

In this case, Eve's result is random relative to Alice's bit. The state resent by Eve gives Bob the correct or incorrect result with equal probability. Therefore,

$$P(\text{error} \mid \text{Eve uses a different basis}) = \frac{1}{2}. \tag{62}$$

The QBER is consequently

$$Q = \frac{1}{2}(0) + \frac{1}{2}\left(\frac{1}{2}\right) \tag{63}$$

$$= \frac{1}{4}. \tag{64}$$

Thus, the intercept–resend attack produces

$$\boxed{Q = 25\%.} \tag{65}$$

**(d) Detecting the attack and completing the protocol**

(i) For $Q = 1/4$ and $m = 20$, the probability of observing at least one error is

$$P(\text{at least one error}) = 1 - (1 - Q)^m \tag{66}$$

$$= 1 - \left(1 - \frac{1}{4}\right)^{20} \tag{67}$$

$$= 1 - \left(\frac{3}{4}\right)^{20} \tag{68}$$

$$\approx 0.9968. \tag{69}$$

Therefore,

$$\boxed{P(\text{at least one error}) \approx 99.68\%.} \tag{70}$$

(ii) The test bits are publicly revealed, so Eve also knows their values. They can no longer be considered secret and must be discarded from the key.

(iii) Error correction allows Alice and Bob to identify and correct disagreements so that they share identical keys. Privacy amplification compresses the corrected key to reduce Eve's possible information about the final key.